

Muskan Pargal

Samba, jammu & kashmir

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Profile

Final-year Electrical Engineering student at Shri Mata Vaishno Devi University with practical exposure to power systems, substations, electrical equipment, protection systems and industrial applications through internships at Hiranagar Grid Station, BHEL and PGCIL Jatwal. Experienced in MATLAB/Simulink and power-electronic project development. Seeking a Semester-long research-oriented internship to apply and strengthen skills in power system protection, renewable energy integration, power electronics and modern grid technologies.

Experience

- POWERGRID- Jatwal Power Grid, 400/220 KV** 22/06/2026 - 22/07/2026
Summer Internship-3
 - Studied the operation and applications of major substation equipment including Circuit Breakers, CTs, CVTs, Isolators, Lightning Arresters, Busbars and Wave Traps.
 - Gained practical exposure to substation protection systems, control and relay panels and SCADA-based monitoring.
 - Studied protection concepts involving Buchholz Relay and Restricted Earth Fault (REF) protection.
 - Observed high-voltage transmission-system operation, switching procedures and protection practices in a 400/220 kV substation.
- BHEL, Haridwar** 04/06/2025 - 01/07/2025
Summer internship-2
 - Gained hands-on exposure to the manufacturing processes of bars and observed practical workflows in various departments.
 - Worked on a project focused on the working and testing of Turbo Generators, understanding their design, operation, and quality checks.
 - Learned about cross-functional processes in manufacturing, testing, and quality assurance, enhancing technical and industrial knowledge.
 - Developed practical insights into large-scale industrial practices, aligning academic concepts with real-world applications.
- Grid Station, Hiranagar** 06/06/2024 - 06/07/2024
Summer Internship-I
 - Learned about the functioning and practical applications of major power system components including Lightning Arrestors (LA), Potential Transformers (PT), Current Transformers (CT), Power Transformers, Circuit Breakers, and Bus Bars.
 - Observed and understood the operation, protection, and switching mechanisms in high-voltage systems.
 - Gained practical insights into grid operation, safety practices, and system reliability.

Projects

- Step-down Chopper Kit**
Arduino-Based PWM Controlled Buck Converter
 - Worked as a team member in designing and developing an Arduino-based PWM controlled step-down (buck) chopper for DC voltage regulation.
 - Implemented PWM generation using Arduino to control duty cycle and regulate output voltage efficiently.
 - Assisted in the design and testing of power circuit using semiconductor devices such as MOSFET/IGBT.
 - Contributed to gate driver circuit integration for high-speed switching performance.
 - Performed experimental analysis and output verification for different duty cycles.
 - Collaborated in hardware setup, debugging, and performance optimization of the system.

Education

- Shri Mata Vaishno Devi University, kakryal, katra** Expected Graduation: 2027
B.tech. in Electrical Engineering
CGPA: 7.76/10

Skills

- Power Systems
- MATLAB/Simulink, Arduino IDE
- Power Electronics